Enhancing Reliability and Efficiency through 24x6 Application Support for Research & Equity Trading

Introduction

A leading global investment management firm, with a strong presence across research, portfolio management, and trading operations, sought a partner capable of delivering round-the-clock production support for its mission-critical applications.

With an expanding footprint across global markets (U.S., U.K., Hong Kong, and Australia), the firm's Research & Equity Trading Group required an experienced managed services partner to provide both Level 1 (L1) and Level 2 (L2) support for a complex suite of financial and analytical applications, including MARKIT EDM (CADIS), FIRE, 4Sight, MAS/BARRA, Minerva, RAP, Sentinel, and more.

The firm engaged NuWare to establish a scalable, 24x6 support model that combined proactive monitoring, issue resolution, and operational governance, ensuring uninterrupted business continuity and compliance with stringent SLAs.

The Challenge

Supporting a distributed and high-volume trading environment meant balancing precision, responsiveness, and system stability. The client faced several key operational challenges:

1. **Complex Multi-Application Ecosystem**
   The Research & Equity Trading Group relied on over 30 interconnected applications; each handling critical processes across portfolio analytics, order management, fixed income risk, and compliance.
2. **High Volume of Batch Jobs**
   The environment processed over 32,000 scheduled jobs daily across the Research and Trading streams, including MARKIT EDM, Minerva, and FIRE, demanding near-zero tolerance for failure.
3. **Demanding SLA and Response Times**
   The client required 24x6 support coverage with five-minute acknowledgment for alerts, as even a minor delay could impact global trading operations.
4. **Manual Monitoring and Coordination Gaps**
   Prior to engagement, there was no unified system for managing user requests, failed batch jobs, and database alerts, leading to delayed escalations.

5. **Audit and Compliance Oversight**
   The firm needed auditable logs and documentation to meet PWC audit requirements and demonstrate operational traceability across trading and research systems.

NuWare's challenge was to deliver a flexible, high-performance support model that could sustain daily operational volume while maintaining SLA precision and proactive issue prevention.

NuWare's Approach

NuWare designed a customized, multi-tiered application support framework combining proactive monitoring, automation, and domain-aligned expertise. The solution ensured end-to-end reliability across trading, research, and portfolio management applications.

1. 24x6 Global Support Operations

- Established an offshore Operations Command Center staffed with 12 dedicated engineers, providing continuous monitoring and support.
- Delivered L1 monitoring and L2 analytical support to ensure uninterrupted system availability across U.S., U.K., HK, and AU time zones.

2. Tiered Support Model

- **L1 Support:**
  - Continuous monitoring of all batch jobs (SOD/EOD checks, file transfers, and report generation).
  - Handling ad-hoc user requests, report scheduling (via SSRS), and data validation tasks (DAA portfolio checks using GO and NAVS).
  - Performing real-time feed checks from external vendors to ensure accuracy and timeliness.
  - Managing access requests, change controls, and Control-M alerts.
- **L2 Support:**
  - Conducting root cause analysis, managing failed SLA batches, and performing transaction-level reconciliation.
  - Executing quality control (QC) checks on accounting transactions, payment workflows, and risk calculation batches.
  - Managing FIT trade application validations and EOD trade file deliveries.
  - Performing annual PWC audit support for trade file QC and data validation.

3. Proactive Monitoring and Escalation

- Automated alert handling for batch job failures, DB blocking, and delayed file feeds.
- Established standard operating procedures (SOPs) for escalation to development teams in case of recurring issues.

4. Continuous Improvement and Documentation

- Maintained process documentation and change logs in Confluence.
- Introduced proactive health checks to identify and prevent recurring issues before impacting production.
- Periodic review meetings with the client to discuss performance metrics, root causes, and enhancements.

Outcomes

The engagement yielded measurable business and operational benefits across reliability, efficiency, and compliance.

1. Enhanced Operational Reliability

- Maintained **99.8% uptime** across all critical trading and research systems.
- Reduced batch job failure rates through proactive issue tracking and optimization.

2. SLA Excellence

- Achieved consistent **five-minute response time** for alerts and incidents.
- Exceeded SLA commitments across ticket closure, batch reruns, and EOD checks.

3. Improved Efficiency and Cost Optimization

- Reduced manual intervention by **40%** through process automation and better coordination.
- Centralized monitoring eliminated the need for regional silos, cutting operational costs.

4. Strong Audit and Compliance Posture

- Provided **fully traceable audit logs** and change management documentation for annual **PWC audits**.

5. Continuous Service Improvement

- Ongoing analysis of repetitive failures led to structural fixes, reducing reoccurrence and improving MTTR (Mean Time to Resolve).

Future Outlook

NuWare continues to partner with the client to evolve the support model with next-generation automation and observability features:

- **Predictive Incident Analytics:** Using AI/ML models to forecast batch failures and SLA breaches.
- **Enhanced Self-Healing Mechanisms:** Auto-restart capabilities for failed jobs and workflows.
- **Intelligent Ticket Routing:** Integration with NLP engines to classify and prioritize incidents automatically.
- **Cloud-Native Monitoring:** Migrating the monitoring ecosystem to a scalable cloud infrastructure for faster alert response and resilience.

These innovations aim to transform the current model into an intelligent operations framework capable of self-monitoring, self-correcting, and scaling dynamically.